Explain HeatShift simply

Presentation and viva rehearsal guide | Slides are required by the latest event notice. Ten minutes includes questions from judges.

Start with this explanation

HeatShift helps people understand tomorrow's temperature and the data behind it. We take weather records, check their quality, estimate the temperature 24 hours ahead and answer supported questions with references. Our dashboard puts all three parts in one place.

The project in six easy steps

1. **Collect:** save original weather files and a receipt saying where each came from.
2. **Organize:** keep raw, cleaned and ready-to-use data separate.
3. **Check:** Mateen's quality agent removes duplicates, fills small gaps and flags unusual values.
4. **Predict:** Waleed's existing model combines a recent airport reading with JMA weather information.
5. **Explain:** Mateen's insight agent retrieves notes and calculates summaries with citations.
6. **Refresh:** an hourly job updates evidence and the dashboard shows whether inputs are available.

What is the model?

HistGradientBoosting combines many small decision trees. Each tree helps correct earlier errors. Our frozen model predicts a correction to the current temperature; adding that correction gives the estimate for the same hour tomorrow. It was already trained before this final audit. Clicking predict does not train it again.

How accurate is it?

MAE means average absolute error. If the actual temperature is 30 C and the forecast is 32 C, the absolute error is 2 C. Across 1,783 reused validation rows, our model's MAE was about 1.10 C, compared with 1.24 C for saying tomorrow equals today. That is 10.8% less average error, not 89.2% accuracy. January was worse than baseline and the uncertainty in improvement includes zero.

Why show a range?

A single number hides uncertainty. March errors give an approximate band around a new prediction: +/-2.22 C in Karachi, +/-3.08 C in Lahore and +/-2.27 C in Multan. These are nominal 90% calibrated ranges, not a promise about future weather. They were not independently tested for future coverage.

What do the two agents actually do?

Quality Agent: acts like a careful spreadsheet reviewer. It checks duplicates, missing values and implausible values, then writes down each decision. Its interpolation can use later values, so cleaned data must be reviewed before modeling.
Insight Agent: acts like a small evidence librarian. It finds relevant sections of trusted notes and summarizes numeric table columns. It adds source row IDs or document section references. It is deterministic retrieval and table reasoning, not a general language-model chatbot. Unsupported AQI and unrelated questions are refused.

What does RAG mean here?

Retrieval-Augmented Generation means finding evidence before constructing an answer. Here retrieval is simple keyword matching over trusted Markdown sections, combined with computed CSV statistics. The added strict question policy accepts whole-table temperature summaries and selected project guidance only. Canonical table hashes and original source row IDs support each number. Dates, other cities, future weather and personal recommendations are refused; arbitrary paraphrases may also be refused. A richer semantic retriever or LLM is future work.

Effort, speed and memory in normal words

Effort: the team's original data/model/agent work is preserved. AI helped substantially with integration, auditing, automation and submission writing. Exact human hours are unknown.
Speed: the model processed 2,538 test inputs in 80.4 milliseconds after loading. The two agents took about 0.24 and 0.10 seconds on 5,012 rows locally. Real clicks also wait for network, downloads and startup.
Memory: the exported model file is about 0.74 MiB; one Node heap snapshot was 11.0 MiB. This is not total browser memory. Browser Python downloads are roughly 25 MB on first use.

What is educational about it?

It teaches data collection receipts, cleaning decisions, time-based splits, honest baseline comparisons, temperature error metrics, calibrated uncertainty, evidence citations and automated updates. It also teaches when to say we do not know.

What is the business model?

Everyday users can use a basic public dashboard. Institutions such as universities or logistics teams could pay for managed views, reports and support. This is a proposed business, not proven sales. Before charging, check data licenses, measure costs and find out whether users will pay.

A three-minute speaking outline

Time	Say and demonstrate
0:00-0:30	The problem, primary user and decision: everyday planning using a three-airport temperature estimate.
0:30-1:15	Explain NOAA/JMA sources, raw-cleaned-processed layers, receipts and fix/fill/flag quality decisions.
1:15-2:15	Walk through the dashboard, model versus baseline, one cited historical answer and one unsupported I do not know.
2:15-3:00	Show city/month errors, January weakness and missing evidence. Explain potential Pakistani planning/data-literacy benefits; impact is unmeasured.

Likely viva questions

Why not random splits? Tomorrow must be predicted using information available earlier; time ordering reduces leakage.
Why compare with a baseline? A complicated model only helps if it improves on a simple rule.
Does it predict all weather? No. The current deployed model predicts temperature at three airports.
What happens when an API fails? The dashboard labels the connection unavailable or stale. It does not invent a successful reading.
Did you improve the model today? We reproduced an existing frozen model's results and added evaluation, calibration and delivery features. No new model fitting happened in this final audit.
What is still missing? Original training-feature/search records, some collection metadata, independent seasonal proof and human review.
Who wrote what? Explain your own contribution and Mateen's agents honestly, using the AI use log. Do not claim AI-assisted code as entirely human-written.

What to submit and bring tomorrow

The organizer explicitly asks for the dataset proposal PDF, one-page summary, AI use log, final repository tag, Space link, public demo under three minutes, business/optimization explanation and all team members at the physical viva. The later event notice explicitly requires slides and a Devpost submission link. Bring the supplied PowerPoint and PDF backup. Historical Prefect evidence is missing; the declaration is late. Do not claim either requirement has been met. This guide is rehearsal material; check any separate organizer announcement for extra rules. Every team member should open the public links and rehearse their own part.

Team contributions

Waleed Irfan developed and improved the model. Syed Mateen Ali developed, managed and maintained both AI agents. Aman Khan Lodhi handled data cleaning, warehouse management and preparation. Roles supplied by Waleed; substantial AI assistance is disclosed separately.